



Heterogeneous change detection via frequency domain interaction and statistical style embedding

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ABSTRACT

Combining the all-weather sensing of Synthetic Aperture Radar (SAR) with optical imagery enables reliable Earth observation, yet their fundamentally different imaging mechanisms cause large representational discrepancies that hinder feature alignment and complicate bi-temporal change detection. To address this issue, we propose FreqHete, a heterogeneous change detection framework that explores both the consistency and disparity between SAR and optical data without auxiliary image generators or adversarial training. First, a style-statistics embedding encoder is developed to inject the deep style statistics of SAR imagery into the low-frequency components of optical features, thereby aligning heterogeneous feature distributions and improving semantic consistency. Second, a frequency-domain interaction decoder is designed to dynamically integrate magnitude and phase information, enabling the model to capture subtle structural variations and enhance sensitivity to change regions. Experiments on the optical-SAR XiongAn benchmark and the satellite-UAV HTCD benchmark show that FreqHete consistently achieves strong overall performance across different heterogeneous change-detection settings while remaining lightweight and efficient. Source code will be released at <https://github.com/weiAI1996/FreqHete>.

1. Introduction

Automatic change detection from remote sensing imagery plays a vital role in understanding Earth surface dynamics [1]. Optical images provide high-resolution observations of surface color, texture, and shape. However, their imaging quality is highly sensitive to environmental conditions. In contrast, Synthetic Aperture Radar (SAR) images reliably capture surface information under all-weather and all-day conditions, which makes SAR particularly valuable in applications such as disaster monitoring, battlefield reconnaissance, and urban management. Driven by these needs, change detection has been actively advanced for a range of application-oriented Earth-observation scenarios in recent years [2,3], spanning high-resolution remote-sensing interpretation [4], semantic land-cover change analysis [5], and heterogeneous multi-source monitoring [6]. By exploiting the complementary strengths of optical and SAR imagery, heterogeneous change detection provides a more comprehensive and accurate characterization of surface changes [7].

Nevertheless, the fundamental differences in imaging mechanisms between SAR and optical data pose significant challenges. Optical imagery depends on reflected light, conveying object color, texture, and shape, but is vulnerable to atmospheric interference, illumination

variations, and cloud occlusion. SAR imagery, on the other hand, is derived from radar backscatter, reflecting surface structures, geometric features, and dielectric properties. As a result, SAR images often exhibit distinct texture characteristics and speckle noise, leading to substantial heterogeneity relative to optical imagery. This divergence complicates the interpretation, alignment, and fusion of the two modalities. In bi-temporal change detection, accurately identifying true change regions while filtering out noise and pseudo-changes caused by modality differences remains a core challenge. Existing homogeneous change detection methods cannot be directly applied to such heterogeneous scenarios, as they often lead to mismatched object representations and erroneous detections.

Early methods for heterogeneous change detection can be categorized into three main approaches: post-classification comparison, similarity measurement, and feature space unification. The post-classification comparison method classifies image pixels from different time periods separately and compares them on a pixel-by-pixel basis to identify changes [8,9]. Similarity measurement methods are based on the assumption that unchanged pixels in heterogeneous images exhibit high similarity within a certain feature space, whereas changed pixels do not, thereby utilizing similarity evaluation for change detection [10].

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Feature space unification methods map heterogeneous images into a unified feature space, ensuring that identical features correspond to the same surface objects, and then compare features across different time points to detect changes [11,12]. In recent years, significant progress has been made in heterogeneous change detection through the use of deep learning techniques for feature space unification. These methods approximate the feature space representation of bi-temporal images using deep networks and perform change detection at either the pixel or object level.

Despite these advances, challenges persist in detecting large-scale building changes in complex urban environments. In particular, fusing bi-temporal heterogeneous data requires not only aligning features across imaging modalities but also disentangling the distinct change patterns unique to optical and SAR data. Addressing these issues necessitates reducing representation discrepancies between modalities while improving the interpretability of heterogeneous change patterns. To this end, this paper introduces a fully supervised framework for heterogeneous change detection that emphasizes cross-modality feature consistency. The proposed framework is designed to accurately capture bi-temporal changes while mitigating the influence of heterogeneity. The main contributions are summarized as follows:

(1) We design a style-statistics embedding encoder that extracts deep style information from SAR images and embeds it into the low-frequency features of optical imagery. This process encourages cross-modality feature distributions to align in a consistent representation space.

(2) We propose a frequency-domain interaction decoder that dynamically integrates heterogeneous features, enabling the extraction of robust difference representations while reducing false detections induced by modality inconsistencies.

(3) Building upon these two components, we develop the FreqHete framework for SAR-optical heterogeneous change detection and further evaluate its frequency-domain alignment principle on an additional satellite-UAV benchmark. Experiments on XiongAn and HTCD demonstrate the effectiveness, efficiency, and cross-sensor robustness of the proposed approach.

2. Related work

2.1. Deep learning-based change detection

With the rapid growth of remote sensing data, the availability of large-scale labeled datasets has driven the advancement of deep learning-based change detection models [5]. Compared with traditional methods, deep learning offers stronger feature representation, capturing both fine details and high-level semantics to handle complex scenes. Existing frameworks are mainly categorized into single-branch and dual-branch structures.

Single-branch models perform early fusion on bi-temporal images through differencing or concatenation, treating change detection as a segmentation task [13–15]. Sun et al. [13] proposed AL-UNet by integrating dilated ConvLSTM into U-Net, enhancing multi-scale spatial modeling. Lin et al. [14] reformulated change detection as video understanding, designing P2V-CD to jointly exploit spatial and temporal dependencies via encoder decoupling and lateral connections.

Dual-branch models adopt late fusion, typically using siamese networks to extract features separately from image pairs [16]. Zhang et al. [16] introduced a deep supervised difference discrimination network with attention modules to refine edge and spatiotemporal features. Lei et al. [17] improved boundary consistency and internal cohesion by modeling long-range relationships. Zhang et al. [18] built a self-structured pyramid network with parallel spatial-channel attention to strengthen multi-scale discrimination in VHR imagery, and Zheng et al. [19] designed a multi-scale hierarchical feature fusion network to progressively aggregate cross-level change cues. To enhance generality, Zhang et al. [4] proposed MCHA-Net, integrating siamese, residual, and

transformer encoders to jointly capture local, global, and cross-domain features.

Transformer models are also widely used to model global context in change detection [20]. Bandara et al. [21] used hierarchical transformers in siamese encoders and reported state-of-the-art results on multiple benchmarks. Chen et al. [22] proposed AGFormer, an anchor-guided transformer that alleviates the severe class imbalance in change detection, while Zhu et al. [23] demonstrated that plain vision transformers (ChangeViT) can serve as strong change detection backbones.

2.2. Heterogeneous change detection

Heterogeneous remote sensing change detection exploits multi-source observations to overcome the limitations of single-modal data. Benefiting from the all-weather imaging capability of SAR, existing studies mainly focus on SAR-optical change detection, and can be broadly grouped into post-classification comparison, similarity measurement, and feature space unification. Post-classification methods compare independently classified multi-temporal images [8,9], but are sensitive to classification errors. Similarity-based methods assume that unchanged regions remain similar across modalities, whereas changed regions exhibit low similarity [10]. Recent work further exploits spatial self-similarity differences to reduce modality-specific noise in heterogeneous images [24]. Feature unification methods instead map heterogeneous data into a shared representation space for direct comparison [11,12]. For example, graph contrastive learning has been introduced to align cross-modal representations through heterogeneous graph contrast [6]. Owing to its robustness, feature-level alignment has become a dominant direction, especially with the development of machine learning and deep learning techniques.

Recent deep models further address SAR-optical discrepancies caused by modality-specific noise and representation gaps. A key challenge is SAR speckle noise, whose statistical behavior and despeckling strategies have been extensively studied [25,26]. This intrinsic radiometric difference makes direct pixel-level SAR-optical comparison unreliable and motivates feature-level alignment. Representative methods include logarithmic transformation for converting multiplicative SAR noise into additive Gaussian-like noise [27], symmetric dual-branch networks for cross-modal feature consistency [28], conditional GAN-based modality projection [29], and siamese transfer networks for shared feature extraction [30]. Recent generative models for heterogeneous change detection include image translation [31,32], reconstruction-based graph autoencoders [33], and denoising diffusion models [34]. Although these methods achieve strong unsupervised performance, they often require auxiliary generators, translators, or iterative samplers. In contrast, FreqHete introduces no generative network and aligns SAR and optical features through a lightweight frequency-domain style embedding. Closest to our supervised setting, HeteCD [35] enforces spatial-domain feature consistency via an auxiliary loss, whereas FreqHete performs architectural alignment in the frequency domain by injecting SAR low-frequency style statistics into optical features.

3. Methodology

In this section, we first introduce the change detection framework FreqHete, which is constructed based on the style-statistics embedding encoder and frequency-domain interaction decoder. Then, two key components of FreqHete are described in detail.

3.1. Overall framework of FreqHete

To mitigate the distributional discrepancy between optical and SAR feature representations and enhance robustness in heterogeneous

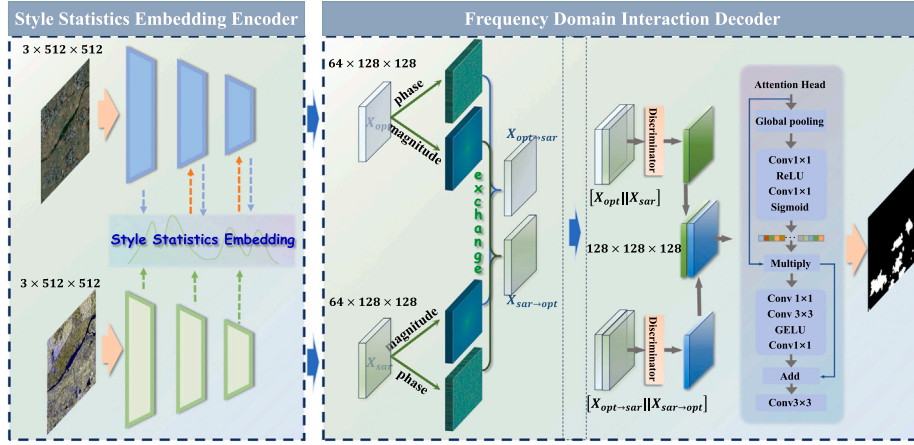


Fig. 1. Overall Framework.

change detection, we propose FreqHete, an end-to-end deep network. FreqHete integrates frequency-domain separation with style-statistics embedding to characterize and align heterogeneous feature distributions. The framework primarily consists of a *style-statistics embedding encoder* and a *frequency-domain interaction decoder*, which jointly capture spatial-temporal consistencies and discrepancies between optical and SAR imagery. The heterogeneous change detection process can be formally expressed as:

$$P = D(E'(I_{opt}), E(I_{sar})), E(I_{sar}), \quad (1)$$

where $D(\cdot)$ denotes the decoder, $E'(\cdot)$ represents the optical encoder augmented with SAR-style statistics, and $E(\cdot)$ is the SAR encoder.

As illustrated in Fig. 1, FreqHete first takes the optical image I_{opt} and SAR image I_{sar} as inputs and extracts multi-scale spatial-temporal features using backbone networks. The SAR features extracted at each hierarchical level are progressively injected into the optical backbone via style-statistics embedding, thereby facilitating feature alignment, style transfer, and distributional consistency. This process is formulated as:

$$\begin{aligned} X_{sar}^{i+1} &= Stage_i(X_{sar}^i), \\ X_{opt}^{i+1} &= Stage'_i(X_{opt}^i, X_{sar}^i), \end{aligned} \quad (2)$$

where $i \in \{0, 1, 2, 3\}$ indexes the encoder depth, $Stage_i(\cdot)$ denotes the i th SAR feature extraction layer, and $Stage'_i(\cdot)$ denotes the i th optical feature extraction layer with SAR-style modulation.

To enhance cross-scale feature integration, FreqHete employs a pyramidal fusion strategy, aggregating multi-level features into a unified high-level representation. This is implemented through 1×1 convolutions and upsampling operations, as follows:

$$X = \mathcal{K}_{1 \times 1}(\Pi_{i=1}^4 \mathcal{K}_{1 \times 1}(U(X_i))), \quad (3)$$

where $\mathcal{K}_{1 \times 1}(\cdot)$ denotes a 1×1 convolution kernel, Π indicates concatenation, and $U(\cdot)$ represents upsampling.

The aggregated optical and SAR features are then fused by a frequency-domain interaction module, yielding bidirectional interaction features $X_{sar \rightarrow opt}$ and $X_{opt \rightarrow sar}$ for modeling cross-modal change patterns. Finally, an attention-based prediction head combines these interaction features with the high-level optical and SAR features (X_{opt} , X_{sar}) to generate a high-resolution change map.

The entire network is trained end-to-end by minimizing a joint objective that linearly combines a class-weighted cross-entropy loss $\mathcal{L}_{ce}$ and a Dice loss $\mathcal{L}_{dice}$:

$$\mathcal{L} = \mathcal{L}_{ce} + \mathcal{L}_{dice}. \quad (4)$$

Let $p_{n,c}$ denote the predicted probability that pixel n belongs to class c ($c \in \{0, 1\}$ for the unchanged and changed classes, respectively) and

$y_{n,c}$ the corresponding one-hot label over the N pixels of a sample. The class-weighted cross-entropy term is

$$\mathcal{L}_{ce} = -\frac{1}{N} \sum_{n=1}^N \sum_{c=0}^1 w_c y_{n,c} \log p_{n,c}, \quad (5)$$

where the class weights $w = [0.2, 0.8]$ are assigned to the unchanged and changed classes to counteract the severe foreground-background imbalance of building changes. The Dice term further alleviates region-level imbalance and is defined on the changed class as

$$\mathcal{L}_{dice} = 1 - \frac{2 \sum_n p_{n,1} y_{n,1} + \epsilon}{\sum_n p_{n,1} + \sum_n y_{n,1} + \epsilon}, \quad (6)$$

where $\epsilon = 10^{-5}$ is a smoothing constant ensuring numerical stability.

3.2. Style-statistics embedding strategy

The texture and noise characteristics of SAR images differ significantly from those of optical images, and these differences are more completely expressed in their distributional statistical information. Deep neural networks can extract feature statistics from SAR images; embedding these statistics into optical features reduces distributional discrepancies and improves cross-modal comparability. Accordingly, we design a style-statistics embedding strategy that injects SAR style statistics into the low-frequency components of optical features during encoding, thereby promoting more consistent heterogeneous feature distributions for change detection. We confine the embedding to the low-frequency band rather than the high-frequency or full spectrum. The low-frequency components encode the global illumination, radiometric mean, and overall tone, which correspond closely to the SAR grayscale statistics and thus constitute the modality-specific style that separates SAR from optical appearance. In contrast, high-frequency components capture fine textures and detailed structures that can vary substantially between modalities; injecting SAR statistics into these components may distort content-bearing structures rather than improve style alignment. Injecting across the full spectrum can also overwrite the spectrum more broadly and introduce phase perturbations that displace structural boundaries and blur genuine changes. Restricting the embedding to the low-frequency band therefore narrows the inter-modal distribution gap while leaving the content-bearing high frequencies intact, consistent with frequency-domain adaptation studies that transfer style through the low-frequency amplitude alone [36].

Specifically, as outlined in Algorithm 1, we begin by applying a Fourier transform to decompose the optical features X_{opt} into frequency components:

$$X_{f,opt} = \mathcal{F}(X_{opt}), \quad (7)$$

Algorithm 1 Style-Statistics Embedding Strategy

Require: X_{opt} , X_{sar} {Input: Feature set X_{opt} (target features), X_{sar} (reference features)}

Ensure: Style-embedded optical feature X_{opt}^{emb}

- 1: Compute the Fourier transform of X_{opt}
- 2: Construct a centered square low-pass mask M_r with default radius ratio $r = 0.25$
- 3: Split the shifted spectrum into low-frequency and high-frequency parts
- 4: Reconstruct the low-frequency optical feature X'_{opt} and compute its statistics
- 5: Compute the mean $\mu_{X_{sar}}$ and standard deviation $\sigma_{X_{sar}}$ of X_{sar}
- 6: Normalize X'_{opt} using its own statistics and the SAR statistics:

$$\hat{X}_{opt}^{low} = \frac{X'_{opt} - \mu_{X'_{opt}}}{\sigma_{X'_{opt}} + \epsilon_s} \cdot \sigma_{X_{sar}} + \mu_{X_{sar}}$$

- 7: Replace only the low-frequency band with $\hat{X}_{opt}^{low}$ and preserve the original high-frequency residual
- 8: **Return** X_{opt}^{emb}

where $\mathcal{F}(\cdot)$ denotes Fourier transform. Next, we shift the low-frequency component to the center of the spectrum through a spectral shift operation:

$$\tilde{X}_{f,opt} = S(X_{f,opt}), \quad (8)$$

where $S(\cdot)$ denotes spectral shift operation. To isolate the low-frequency components, a centered square mask M_r is constructed to occlude the high-frequency regions while preserving the central low-frequency region:

$$\tilde{X}_{f,low,opt} = \tilde{X}_{f,opt} \odot M_r, \quad \tilde{X}_{f,high,opt} = \tilde{X}_{f,opt} \odot (1 - M_r), \quad (9)$$

where $\odot$ denotes point-wise multiplication. For the default FreqHete setting, M_r is shared across channels and set to one inside the centered square region $|u - H/2| \leq rH$ and $|v - W/2| \leq rW$ after spectral shift, and zero elsewhere. We use $r = 0.25$ unless otherwise specified, which retains $(2r)^2 = 25\%$ of the Fourier coefficients. The low-frequency component is then subjected to a spectral inverse shift operation to relocate the central low-frequency information to the periphery. Subsequently, the inverse Fourier transform is applied to reconstruct the spatial-domain content image X'_{opt} :

$$X'_{opt} = i\mathcal{F}(iS(\tilde{X}_{f,low,opt})), \quad (10)$$

where $i\mathcal{F}(\cdot)$ denotes inverse Fourier transform and where $iS(\cdot)$ denotes spectral inverse shift operation. Additionally, to extract statistical guidance from SAR image features and normalize the low-frequency optical feature, we calculate channel-wise instance statistics over the N spatial positions of each sample:

$$\begin{aligned} \mu_{X_{sar}} &= \frac{1}{N} \sum_{i=1}^N X_{sar}^{(i)}, \\ \sigma_{X_{sar}}^2 &= \frac{1}{N} \sum_{i=1}^N (X_{sar}^{(i)} - \mu_{X_{sar}})^2, \\ \mu_{X'_{opt}} &= \frac{1}{N} \sum_{i=1}^N X'_{opt(i)}, \\ \sigma_{X'_{opt}}^2 &= \frac{1}{N} \sum_{i=1}^N (X'_{opt(i)} - \mu_{X'_{opt}})^2. \end{aligned} \quad (11)$$

With the computed statistics from X_{sar} , we normalize the low-frequency components X'_{opt} by applying the instance normalization technique,

adjusting the mean and variance of X'_{opt} to be consistent with the mean and variance of X_{sar} :

$$\hat{X}_{opt}^{low} = \frac{X'_{opt} - \mu_{X'_{opt}}}{\sigma_{X'_{opt}} + \epsilon_s} \cdot \sigma_{X_{sar}} + \mu_{X_{sar}}, \quad (12)$$

where ϵ_s is a small constant for numerical stability. The normalized low-frequency feature is then transformed back to the frequency domain and used to replace only the low-frequency band, while the original high-frequency optical residual is preserved:

$$\begin{aligned} \hat{X}_{f,low,opt} &= S(\mathcal{F}(\hat{X}_{opt}^{low})) \odot M_r, \\ X_{opt}^{emb} &= i\mathcal{F}(iS(\hat{X}_{f,low,opt} + \tilde{X}_{f,high,opt})). \end{aligned} \quad (13)$$

Embedding SAR style statistics into optical features in this manner reduces the distributional gap between heterogeneous images while keeping the content-bearing high-frequency optical details intact, facilitating feature alignment for change detection.

3.3. Frequency-domain interaction decoder

Although the style-statistics embedding encoder narrows the heterogeneity gap in the feature space, it remains challenging to eliminate such discrepancies completely. To further improve the model's capacity for discriminating land-cover changes, we propose a decoding strategy that explicitly leverages frequency-domain interactions.

Specifically, heterogeneous features are decomposed into magnitude and phase components through Fourier transform, where the magnitude primarily reflects style-related information (e.g., texture intensity and radiometric properties), while the phase captures content-related structures (e.g., spatial layouts and object boundaries). This decomposition provides a natural separation between style and content cues. This property is particularly pertinent to the heterogeneous SAR-optical setting: after co-registration, unchanged or weakly changed regions share the same ground geometry across the two modalities, and the positions of many edges and object boundaries encoded by the phase spectrum are therefore more consistent than their radiometric appearances, whereas the magnitude spectrum is governed by sensor-specific scattering mechanisms and radiometric characteristics. Consequently, the inter-modal discrepancy is concentrated mainly in the amplitude, so exchanging magnitudes while preserving phase transfers modality-specific style without perturbing the shared structural layout in these regions. This rationale is consistent with classical and frequency-domain style-transfer studies showing that phase encodes image structure while amplitude encodes appearance and style [36]. By exchanging and recombining the magnitude and phase components of optical and SAR features, the model can explicitly simulate cross-modal style-content interactions, thereby enriching the diversity of feature representations and reducing modality bias. This operation encourages the model to learn change-relevant information that remains consistent under style perturbations, thereby improving robustness and generalization. We further examine this separation on the XiongAn features entering the decoder: as shown in Fig. 2, the magnitude spectra of the optical and SAR features are clearly separated (Wasserstein-1 distance $W_1 = 0.80$), whereas the phase spectra have a much smaller distributional gap ($W_1 = 0.012$). This empirical evidence supports the interpretation that the inter-modal discrepancy is mainly reflected in the amplitude statistics, while the phase-related structural distributions are more consistent across modalities, justifying our choice of exchanging magnitudes while preserving phase.

As shown in Fig. 1, the encoder outputs X_{opt} and X_{sar} , corresponding to optical and SAR modalities, are transformed into frequency domain components. Mixed features are then generated by swapping magnitude and phase between modalities as follows:

$$\begin{aligned} X_{opt \rightarrow sar} &= i\mathcal{F}(A_{sar}(u, v)e^{j\phi_{opt}(u, v)}), \\ X_{sar \rightarrow opt} &= i\mathcal{F}(A_{opt}(u, v)e^{j\phi_{sar}(u, v)}), \end{aligned} \quad (14)$$

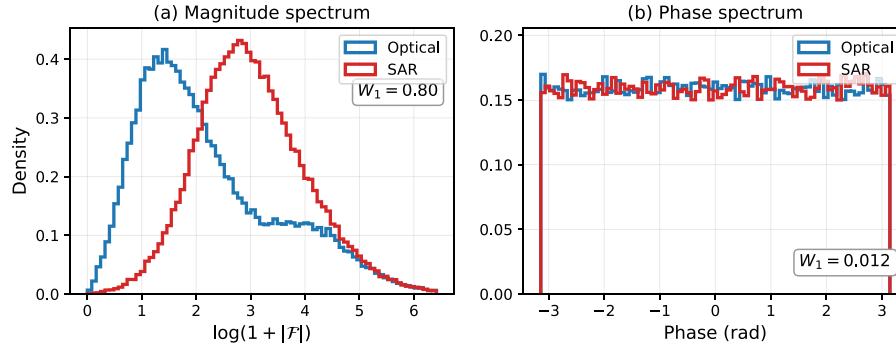


Fig. 2. Distributions of the (a) magnitude and (b) phase spectra of the optical and SAR deep features entering the frequency-domain interaction decoder, aggregated over the XiongAn evaluation set. The magnitude distributions of the two modalities are clearly separated (Wasserstein-1 distance $W_1=0.80$), whereas their phase distributions show a much smaller gap ($W_1=0.012$). This empirical observation supports our design: the inter-modal style discrepancy is mainly reflected in the magnitude statistics, while the phase-related structural distributions are more consistent across modalities.

where $A(u, v)$ denotes the magnitude spectrum and $e^{j\phi(u, v)}$ denotes the phase. The resulting features combine the complementary strengths of both modalities, compelling the network to attend to structural consistencies (phase) while being invariant to style discrepancies (magnitude).

To further capture discriminative cues, a difference discriminator $D(\cdot)$ is introduced to highlight discrepancies between the swapped and original features:

$$X_{diff} = D(X_{opt} \parallel X_{sar}) \parallel D(X_{opt \rightarrow sar} \parallel X_{sar \rightarrow opt}). \quad (15)$$

The attention mechanism is then applied to emphasize informative change regions:

$$\tilde{X}_{diff} = X_{diff} \odot \sigma(\mathcal{K}_{1 \times 1}(\delta(\mathcal{K}_{1 \times 1}(\text{GAP}(X_{diff}))))), \quad (16)$$

where σ is the sigmoid function, δ is ReLU, and $\text{GAP}(\cdot)$ denotes global average pooling. Finally, a residual prediction head generates the change detection map.

By exploiting magnitude-phase exchanges, the proposed decoder reduces modality-specific differences and improves sensitivity to structural variations in heterogeneous change detection.

4. Experiments

In this section, we first evaluate FreqHete on XiongAn optical-SAR pairs and then conduct an additional study on the satellite-UAV HTCD benchmark.

4.1. Experimental protocol

(1) Xiong'an dataset: The heterogeneous data were collected from the Xiong'an New Area in China, which underwent extensive urban development between 2018 and 2023. The Xiong'an dataset [35] consists of pre-event multispectral optical images and post-event quad-polarimetric SAR images acquired in StripMap mode, containing a large number of building change instances. The optical images were captured by the GF-2 satellite with a spatial resolution of 4 m, while the SAR images were acquired by the GF-3 satellite with a spatial resolution of 8 m. To make the heterogeneous modalities directly comparable, we adopt the alignment and preprocessing pipeline released with the dataset benchmark [35]. Specifically, the SAR images are first multi-looked and despeckled using a refined Lee filter to suppress speckle noise while preserving structural details. Subsequently, both modalities are geocoded and radiometrically calibrated into a unified geographic coordinate system. A set of manually selected tie points in the overlapping regions is then employed to co-register the bi-temporal optical and SAR scenes and resample them onto a shared spatial grid, ensuring that corresponding pixels represent the same ground location. For network

input construction, the red, green, and blue bands of the optical images together with the HH, VV, and HV polarizations of the SAR images are retained. The co-registered image pairs are cropped into 512×512 patches with a stride of 256. Following the benchmark release, the resulting patch tensors are stored in an 8-bit image range after the above calibration and alignment steps, and are normalized to $[0, 1]$ by dividing by 255 before being fed to the network. The dataset contains 1901 image pairs for training and 413 pairs for testing. To improve model generalization, random data augmentation strategies, including random cropping, flipping, color space transformation, and rotation, are applied during training.

(2) HTCD dataset [37]: The HTCD dataset is designed for heterogeneous change detection using satellite and unmanned aerial vehicle (UAV) imagery. It contains bi-temporal data acquired between 2008 and 2020, covering approximately 36 km² around Kishinev and its surrounding areas. The dataset mainly focuses on urban changes, such as modifications to buildings and roads, with man-made objects carefully annotated, while natural changes are excluded. The satellite images have a spatial resolution of 0.5971 m, whereas the UAV images provide a higher resolution of 7.465 cm. During training, the images are cropped into 256×256 patches, resulting in 3080 training pairs and 1321 testing pairs. Other experimental settings are kept consistent with those used for the Xiong'an dataset.

(3) Experimental Setting: All experiments were conducted on a server running Ubuntu 18.04 and equipped with eight NVIDIA RTX A6000 GPUs, each with 48 GB of VRAM. The implementation was based on PyTorch 1.10.0 with CUDA 11.3 and cuDNN 8.2. Unless otherwise required by a released baseline implementation, the initial learning rate was set to 0.0002 and updated using a linear decay schedule, the batch size was set to 8 during training and 1 during testing, and the model was trained for 400 epochs using the AdamW optimizer. For baselines with implementation-specific optimizer, initialization, or architecture settings, we retained the public configuration while using the same dataset splits, input protocol, and evaluation metrics. For reproducibility, methods with standard CNN or Transformer backbones used ImageNet pretrained weights when this initialization was supported by the released implementation and benchmark protocol [35]; methods without an explicit backbone or with fixed released initializations followed their public configurations. To ensure a fair comparison, no additional training data, test-time augmentation, or task-specific tuning beyond the public implementations was introduced.

4.2. Comparison with advanced methods

To evaluate the effectiveness of FreqHete, we compare it with 14 state-of-the-art change detection methods. The compared methods include STANet [38], IFN [16], DTCDSN [39], SNUNet [40],

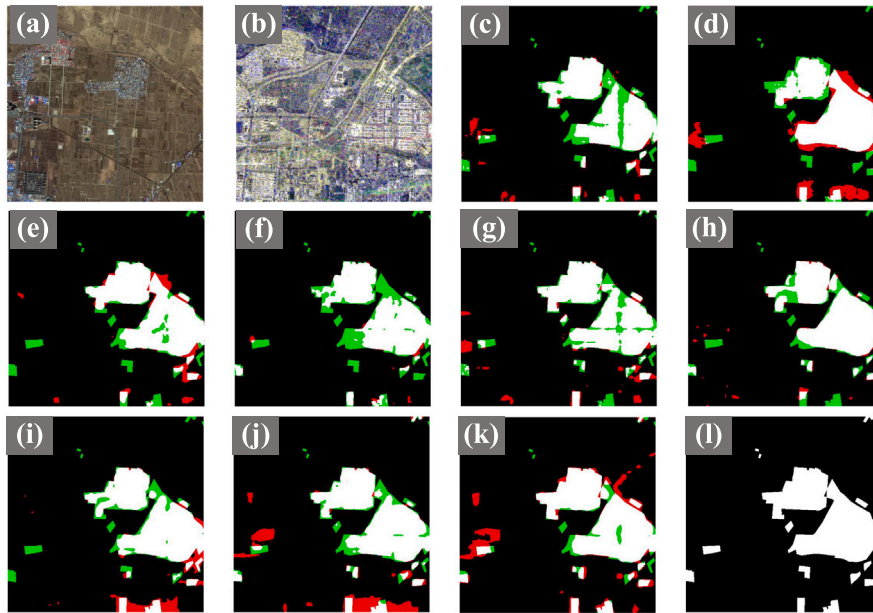


Fig. 3. The qualitative comparison of various methods in the context of high-rise residential construction scenarios. (a) Pre-temporal image. (b) Post-temporal image. (c) ChangeFormer. (d) HANet. (e) LightCDNet. (f) CGNet. (g) Changer. (h) TTP. (i) BAN. (j) HeteCD. (k) FreqHete. (l) Ground truth. We highlight the TP areas in white, the FP areas in red, and the FN areas in green. Black denotes the TN areas. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

Quantitative Results on the XiongAn. The best results are marked in bold. FPS is measured at 512×512 input resolution with batch size 1 in FP32 on a single NVIDIA RTX A6000 GPU.

Method	Backbone	Publication	Params (M)	MACs (G)	FPS	Metrics (%)			
						Precision	Recall	F1	IoU
STANet	ResNet18	RS 2020	13.36	500.36	9.1	56.31	61.24	58.67	41.52
IFN	VGG16	ISPRS 2020	36.00	315.93	21.4	80.49	80.70	80.59	67.50
DTCDSN	SE-ResNet18	GRSL 2021	31.26	52.90	68.3	83.20	69.44	75.71	60.91
SUNet	UNet	GRSL 2021	12.04	186.79	17.2	80.85	63.85	71.35	55.46
ChangeStar	ResNet18	ICCV 2021	16.97	76.85	50.4	85.50	75.24	80.04	66.72
BIT	ResNet18	TGRS 2021	2.99	35.00	57.8	75.35	70.05	72.60	56.99
ChangeFormer	MiT-b1	arXiv 2022	41.03	811.15	8.6	84.24	79.94	82.04	69.54
HANet	ResNet18	JSTARS 2023	3.02	97.55	23.9	76.46	52.36	62.16	45.09
LightCDNet	–	GRSL 2023	2.82	23.80	31.0	80.24	72.81	76.34	61.74
CGNet	–	JSTARS 2023	38.99	415.88	24.3	90.11	74.24	81.41	68.65
Changer	ResNet18	TGRS 2023	11.39	23.82	74.1	82.26	78.92	80.56	67.44
TTP	SAM-l	IGARSS 2024	352.49	908.04	6.5	84.63	64.34	73.10	57.61
BAN	Vit-b	TGRS 2024	68.23	65.41	25.0	82.05	79.94	80.99	68.05
HeteCD	MiT-b1	ISPRS 2025	50.19	58.23	24.0	81.18	80.02	80.60	67.50
FreqHete	ResNet18	–	22.95	37.95	90.6	81.31	85.28	83.25	71.30

ChangeStar [41], BIT [42], ChangeFormer [21], HANet [43], LightCDNet [44], CGNet [45], Changer [46], TTP [47], BAN [48], and HeteCD [35]. Because publicly available fully supervised baselines tailored to heterogeneous optical-SAR change detection remain limited, we include established homogeneous change detection methods adapted to heterogeneous inputs, along with representative heterogeneous and foundation-model-based approaches, under the controlled protocol described above.

Quantitative Comparison: We use Intersection over Union (IoU), Precision, Recall, and F1-score (F1) to quantitatively evaluate change detection performance. The quantitative comparison results are summarized in Table 1. FreqHete achieves the highest Recall (85.28%), F1-score (83.25%), and IoU (71.30%) among the compared methods. These results indicate that FreqHete captures change regions accurately and improves overall change detection performance. In terms of precision, FreqHete does not exhibit a significant advantage over other methods. The qualitative analysis indicates that some false detections occur in regions with disputed annotations and in high-intensity SAR areas. In terms of efficiency, FreqHete has 22.95M parameters and

37.95G MACs. These costs are lower than those of ChangeFormer (41.03M, 811.15G) and TTP (352.49M, 908.04G), indicating a compact computational profile. Compared with models of similar parameter size, such as Changer, FreqHete achieves higher recall, F1-score, and IoU while maintaining competitive precision. Overall, FreqHete achieves a favorable balance between accuracy and efficiency among the compared methods. FreqHete also attains the highest inference speed among the compared methods, reaching 90.6 FPS at 512×512 resolution with batch size 1 on a single RTX A6000 GPU, compared with LightCDNet (31.0 FPS), STANet (9.1 FPS), ChangeFormer (8.6 FPS), and TTP (6.5 FPS). Although FreqHete has neither the fewest parameters nor the lowest MACs, its computation is concentrated in dense convolutions over low-resolution feature maps, resulting in a small activation footprint and high GPU utilization. This suggests that the additional Fourier transforms and frequency-domain interaction introduce limited practical overhead.

Qualitative Comparison: Fig. 3 illustrates the transformation of rural low-rise residential areas into high-rise buildings, as well as the

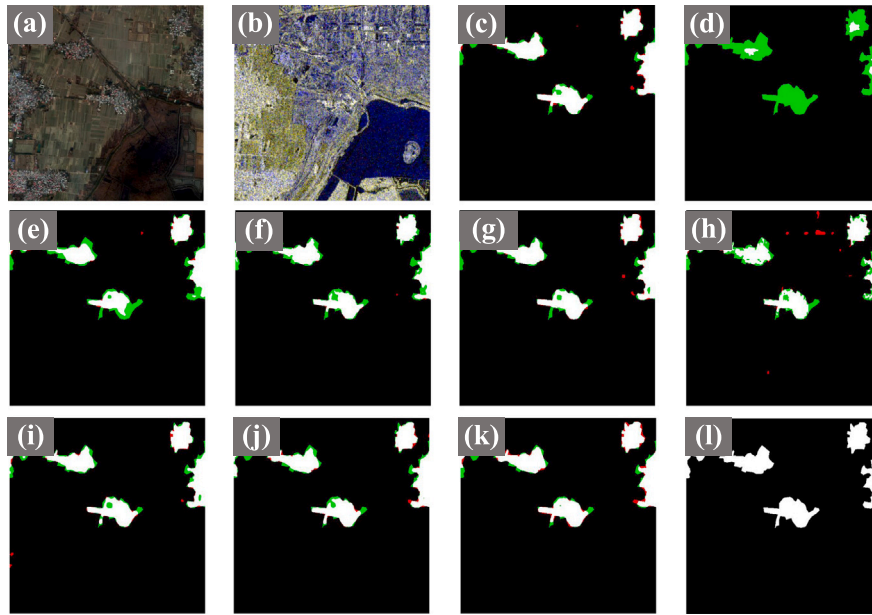


Fig. 4. The qualitative comparison of various methods in the context of low-rise town demolition scenarios. (a) Pre-temporal image. (b) Post-temporal image. (c) ChangeFormer. (d) HANet. (e) LightCDNet. (f) CGNet. (g) Changer. (h) TTP. (i) BAN. (j) HeteCD. (k) FreqHete. (l) Ground truth. We highlight the TP areas in white, the FP areas in red, and the FN areas in green. Black denotes the TN areas. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2

Quantitative Results on the HTCD dataset. The best results are marked in bold.

Method	Precision	Recall	F1	IoU
STANet	51.31	72.38	60.05	42.91
IFN	89.96	91.13	90.54	82.72
DTCDCSCN	94.78	80.13	86.84	76.74
SNUNet	63.09	71.74	67.14	50.53
BIT	47.74	87.34	61.74	44.65
ChangeFormer	94.61	90.09	92.30	85.70
HANet	69.91	73.22	71.53	55.67
CGNet	92.58	89.27	90.90	83.31
Changer	90.45	87.80	89.11	80.36
TTP	93.46	90.84	92.13	85.41
BAN	78.78	85.77	82.13	69.68
HeteCD	94.21	94.49	94.34	89.30
FreqHete	93.13	97.22	95.13	90.72

addition of new structures in previously bare land regions. Several compared methods miss small-scale newly added buildings. In contrast, the proposed method detects most annotated building changes, although false detections remain in regions with disputed changes and high-intensity SAR responses. Moreover, FreqHete preserves finer boundary and small-object details than the compared methods in this example. Fig. 4 depicts a scenario where rural buildings have been demolished, the foundations have been laid, but high-rise construction has not yet commenced. The change detection results from optical and SAR images reveal that HANet exhibits omissions in localized areas, while other methods suffer from varying degrees of detail omission or false detections. In comparison, the proposed method detects finer details more consistently than the compared methods. Fig. 5 presents a scene containing extensive areas of unreconstructed buildings and changes in low-rise structures within agricultural fields. In this scenario, methods such as HANet, LightCDNet, and CGNet miss many of these changes, while other approaches show varying degrees of omissions or false detections near building edges. Overall, the proposed method performs favorably among the compared techniques in detail preservation and change detection accuracy.

4.3. Generalization on the HTCD dataset

Quantitative Comparison: To examine transfer beyond the optical-SAR setting of XiongAn, we further evaluate FreqHete on HTCD, which involves satellite-UAV heterogeneity with large platform, viewpoint, and resolution gaps. As summarized in Table 2, FreqHete obtains the highest Recall (97.22%), F1-score (95.13%), and IoU (90.72%) among all compared methods. It exceeds the dedicated heterogeneous baseline HeteCD by 1.42 percentage points in IoU (90.72% vs. 89.30%) and achieves higher IoU than ChangeFormer (85.70%) and TTP (85.41%). Although DTCDCSCN attains a marginally higher precision, its recall (80.13%) and IoU (76.74%) are substantially lower, whereas FreqHete maintains a better precision-recall balance. These results are consistent with the XiongAn observations and suggest promising transfer across different source-heterogeneity settings. For HTCD, the second modality is treated as the reference modality for style-statistics alignment, so the SAR-specific formulation used for XiongAn is instantiated as a general reference-modality style embedding.

4.4. Ablation studies

We ablate the style-statistics embedding strategy and the frequency-domain interaction decoder on the XiongAn dataset by removing specific components from FreqHete. The quantitative results are provided in Table 3.

(1) *Baseline:* The baseline network employs ResNet18 as the backbone, with the optical and SAR branches independently optimized during the feature extraction process, without any interaction between them. For difference feature modeling, the backbone features are first concatenated and then processed using 2D convolution to extract difference features. The frequency-domain interaction decoder is replaced with an FFN decoder. To ensure a fair comparison, all other hyperparameters are kept consistent across all comparative experiments. The baseline achieves an F1 score of 80.41% and an IoU of 67.24% on the XiongAn dataset.

(2) *Effects of style-statistics embedding strategy:* To address the significant representational differences between optical and SAR images,

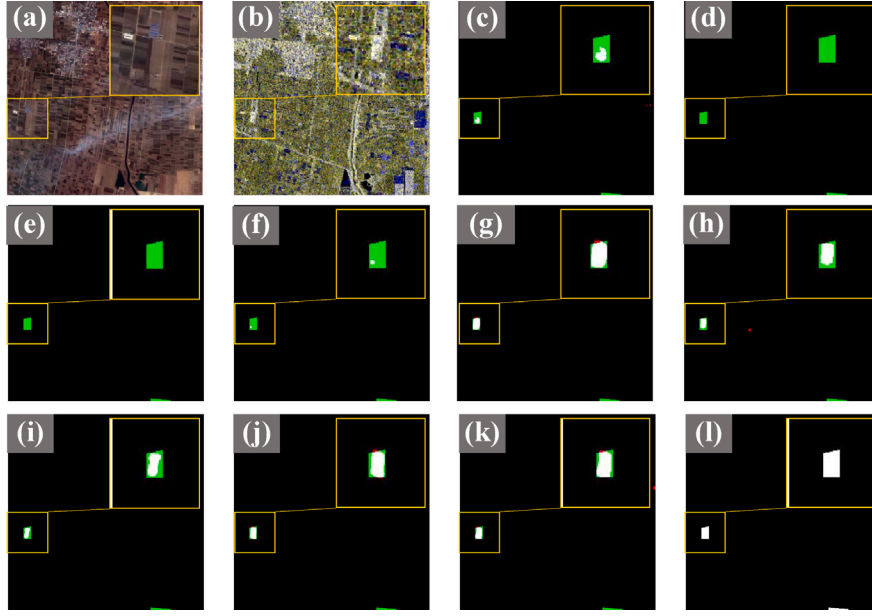


Fig. 5. The qualitative comparison of various methods in the context of small standalone building change scenarios. The yellow box marks the small building change region, which is magnified in the top-right inset of each sub-figure for clearer comparison. (a) Pre-temporal image. (b) Post-temporal image. (c) ChangeFormer. (d) HANet. (e) LightCDNet. (f) CGNet. (g) Changer. (h) TTP. (i) BAN. (j) HeteCD. (k) FreqHete. (l) Ground truth. We highlight the TP areas in white, the FP areas in red, and the FN areas in green. Black denotes the TN areas. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3

Ablation experiment on the XiongAn dataset. The best results are marked in bold.

Style-Statistics Embedding Strategy	Frequency-Domain Interaction Decoder	Precision	Recall	F1	IoU	Params	MACs
		72.89	89.66	80.41	67.24	22.89	36.56
✓		78.71	85.51	81.97	69.44	22.89	36.56
	✓	77.44	86.65	81.78	69.19	22.95	37.95
✓	✓	81.31	85.28	83.25	71.30	22.95	37.95

we propose a style-statistics embedding strategy to model the commonalities and uniqueness of deep semantic features in heterogeneous images. As shown in Table 3, the comparative experiments (e.g., Rows 2 and 4) show the performance improvement brought by this module. Specifically, integrating the style-statistics embedding strategy into the baseline model results in a 2.2 percentage-point increase in the change-class IoU metric, improving from 67.24% to 69.44%. This result indicates that the style-statistics embedding strategy improves semantic feature alignment in heterogeneous images.

(3) *Effects of frequency-domain interaction:* To improve change representation in heterogeneous images, we design a frequency-domain interaction decoder for feature interaction and fusion in the frequency domain. As evidenced in Table 3, the ablation experiments on different component combinations support the effectiveness of this decoder. Its standalone integration leads to a 1.95 percentage-point improvement in change-class IoU. When combined with the style-statistics embedding strategy, the frequency-domain interaction decoder further improves FreqHete's performance. Compared to the baseline model, it delivers a 4.06 percentage-point increase in change-class IoU and a 2.84 percentage-point increase in the F1 score. These results further support the effectiveness of the proposed approach in modeling representational differences and detecting changes in heterogeneous images.

To further assess the effectiveness of frequency-domain interaction in extracting robust difference features, Gradient-weighted Class Activation Mapping (Grad-CAM) was employed to visually compare

network layers with and without frequency-domain interaction. Without frequency-domain interaction, the layers rely mainly on the original heterogeneous features to construct the difference representation, which may lead to incomplete representation or noise interference. As shown in Fig. 6, the first-row visualization suggests that the proposed frequency-domain interaction strategy covers change regions more completely and improves sensitivity to complex scene changes. Additionally, in the second row, the gradient maps generated by network layers without frequency-domain interaction exhibit noticeable noise, manifested as spurious activations in irrelevant regions. With frequency-domain interaction, the Grad-CAM responses become more concentrated on annotated change regions, indicating reduced interference from irrelevant activations. The third row highlights the effect of frequency-domain interaction in extracting features of small change targets, such as fine-grained structures like small buildings. The frequency-domain interaction strategy improves the model's focus on local changes and reduces false detections associated with inconsistent feature distributions. This result supports the contribution of frequency-domain interaction to modeling heterogeneous feature differences and improving change detection in challenging scenarios.

(4) *Analysis of frequency-domain design parameters:* We ablate three frequency-domain choices in FreqHete's style-statistics embedding: the carrier *band*, the low-pass mask radius *r*, and the mask *shape*. All variants are trained for 150 epochs under the same controlled budget for this sensitivity study. Results are reported in Table 4 and Fig. 7; band selection has the largest effect. Replacing the low-pass band with

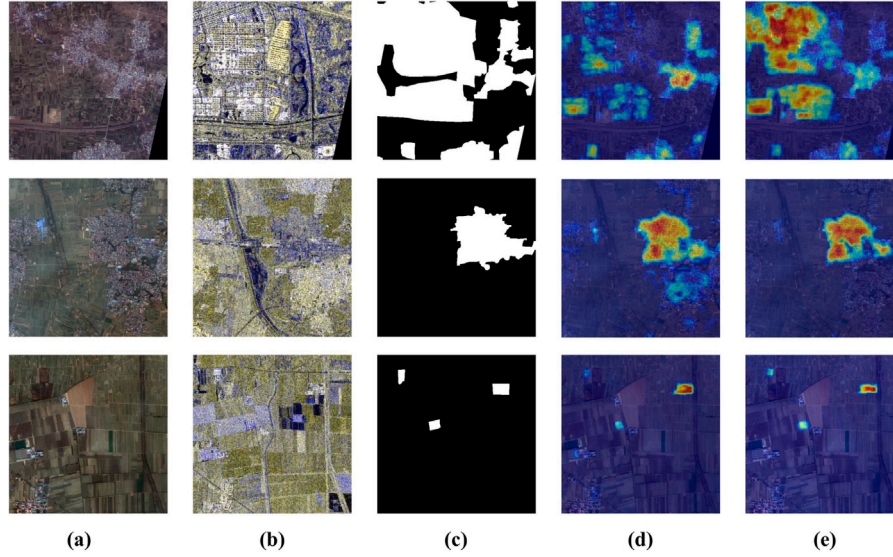


Fig. 6. Example of frequency-domain interaction visualization by Gradient-weighted class activation maps (Grad-CAM). (a) Pre-temporal image. (b) Post-temporal image. (c) Ground truth. (d) Grad-CAM of the model without frequency-domain interaction. (e) Grad-CAM of the model with frequency-domain interaction.

Table 4

Sensitivity analysis of frequency-domain design parameters on XiongAn. All variants use the same 150-epoch controlled budget. The default low-pass square mask with $r=0.25$ obtains 69.75% change-IoU, close to the 71.30% under the full 400-epoch setting. Δ IoU reports the paired per-image bootstrap difference from the default in percentage points with 95% confidence intervals; **bold** indicates intervals that exclude zero.

Setting	Retained coeff. (%)	Change-IoU (%)	Δ IoU (pp) [95% CI]
<i>(a) Frequency band</i>			
low-pass ($r=0.25$, default)	25	69.75	–
full spectrum	100	69.32	–0.43 [–1.2, +0.4]
mid annulus	14.5	65.96	–3.79 [–4.9, –2.6]
high-pass	75	65.06	–4.69 [–5.9, –3.5]
<i>(b) Low-pass mask size</i>			
$r=0.10$	4	68.80	–0.95 [–1.8, –0.2]
$r=0.25$ (default)	25	69.75	–
$r=0.40$	64	69.55	–0.20 [–0.9, +0.5]
<i>(c) Mask shape (bandwidth-matched)</i>			
square (default)	25	69.75	–
circle	25	69.43	–0.32 [–1.1, +0.4]
gaussian	25	68.84	–0.91 [–1.8, –0.01]

a high-pass band (–4.69) or a mid-frequency annulus (–3.79) causes substantial degradation, since SAR speckle and optical texture mainly reside in modality-specific high-frequency components, whose transfer can disrupt content structures. Full-spectrum injection also brings no clear gain over the low-pass design, despite using about four times more Fourier coefficients. Mask size and shape have smaller effects. The tested radii $r = 0.25$ and $r = 0.40$, as well as a bandwidth-matched circular mask, yield statistically comparable results, whereas an overly small mask ($r = 0.10$) and a soft Gaussian profile degrade performance. Overall, the low-pass/square/ $r = 0.25$ setting achieves the highest mean IoU among all variants, supporting the use of low-frequency embedding in FreqHete.

(5) *Robustness via evaluation-level five-fold validation:* To assess the stability of the reported accuracy across data partitions, we perform an evaluation-level five-fold validation over the held-out 413-tile XiongAn evaluation set. The change-class IoU across the five disjoint folds is 69.1, 70.9, 64.1, 72.7 and 70.0% (mean $69.4 \pm 3.2\%$), and the F1 is $81.9 \pm 2.3\%$; averaged over 200 random fold partitions, the fold-wise standard deviation remains 0.038 in IoU and 0.027 in F1, i.e. a coefficient of variation of only 5.5% and 3.3%, respectively. This small dispersion, together with the per-image bootstrap confidence intervals reported above, indicates that FreqHete’s accuracy is stable across data

partitions. Overall, the evaluation-level five-fold validation provides quantitative evidence for stable performance across different held-out subsets.

4.5. Failure case analysis

Although FreqHete achieves the highest Recall, F1-score, and IoU among the compared methods in Table 1, it still produces a small number of false positives, visible as the red regions in Figs. 3–5. We observe that these errors concentrate in two specific situations and analyze their underlying causes below. First, false positives tend to occur in *high-intensity SAR regions*. Strong-backscatter surfaces – such as metallic roofs, dihedral structures, and densely built-up areas – yield abnormally large energy, and hence large mean and variance, in their deep SAR features. Because our style-statistics embedding injects the mean and variance of the SAR features into the low-frequency components of the optical features (Algorithm 1), these anomalous statistics are occasionally transferred as a spurious “style”, shifting the optical feature distribution in those areas and triggering change-like responses even where no genuine change exists. This is therefore primarily a *modality-induced* side effect of the alignment mechanism rather than a generic classification failure, and it is intrinsically tied

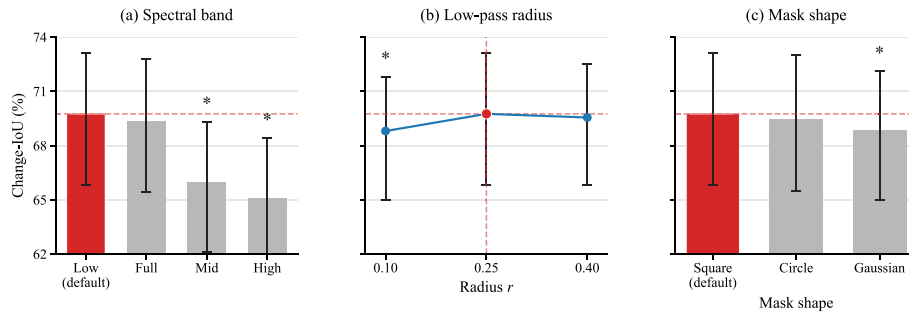


Fig. 7. Frequency-domain parameter sensitivity on XiongAn: (a) spectral band, (b) low-pass radius, and (c) mask shape. Bars/markers show change-IoU with per-image bootstrap 95% confidence intervals. Red denotes the default setting; asterisks mark variants whose paired-bootstrap Δ IoU from the default excludes zero, with exact intervals reported in Table 4. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

to our low-frequency injection design. Second, false positives appear in *regions with disputed annotations*. Building changes frequently involve transitional construction states – for example, sites where the old structures have been demolished and the foundations laid but the superstructure has not yet been built (Fig. 4) – whose change status is inherently ambiguous and is labeled inconsistently in the ground truth. The false positives in these regions may partly reflect *label noise*, which is consistent with the comparatively lower precision and high recall of FreqHete reported in Table 1.

These observations point to two architecture-level directions for mitigation. (i) Replacing the fixed low-frequency mask with *content-adaptive, learnable spectral selection* could allow the model to down-weight unreliable high-energy SAR bands in strong-backscatter regions. (ii) Incorporating *uncertainty modeling* into the prediction head would enable the network to flag and attenuate low-confidence responses in ambiguously annotated areas.

5. Conclusion

In this paper, we present FreqHete, a frequency-domain framework designed for SAR-optical heterogeneous change detection and further evaluated in a satellite-UAV heterogeneous setting. The method aligns cross-modal features by injecting the style statistics of the reference modality into low-frequency components while preserving structural high-frequency cues, and it employs a magnitude-phase interaction decoder to disentangle style-dominated magnitude from structure-dominated phase. Experiments on XiongAn and HTCD show that this design improves change-region detection under both heterogeneous settings while remaining lightweight, generator-free, and end-to-end trainable.

FreqHete provides a simple frequency-domain pathway for handling heterogeneous observations when sensors share scene geometry but differ markedly in radiometry, texture, and noise. The proposed style-statistics embedding and magnitude-phase interaction can serve as plug-in modules for existing encoders, giving researchers a flexible tool, and offer practitioners a compact alternative to image-translation or adversarial alignment pipelines. Failure analysis further clarifies the method's behavior: high-intensity SAR responses and ambiguously annotated construction states remain the main sources of false positives, highlighting where uncertainty-aware prediction or adaptive spectral selection would be most beneficial.

We also acknowledge three limitations. First, the low-frequency mask is fixed rather than content-adaptive, so it cannot yet suppress unreliable spectral bands on a per-scene basis. Second, validation is conducted on public benchmarks rather than operational deployment data, leaving real-world robustness under varying acquisition conditions to be further examined. Third, the framework currently focuses on bi-temporal binary change detection. In future work, we will replace the fixed mask with content-adaptive spectral selection, extend

FreqHete to multi-temporal and semantic change detection, evaluate additional sensor pairs such as SAR-hyperspectral and SAR-LiDAR, and integrate the method as a frequency-domain front-end for multimodal remote sensing foundation models.

CRediT authorship contribution statement

Wei Jing: Writing – review & editing, Writing – original draft. **Qiang Li:** Validation. **Qi Wang:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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